



FRONT-END WEB DEVELOPMENT

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GIT



WHAT IS VERSION CONTROL?

A system that keeps records of your changes

Allows for collaborative development

Allows you to know who made what changes and when

Allows you to revert any changes and go back to a previous state



WHAT IS GIT?

- ❑ Started in 2005
- ❑ Created by Linus Torvald to aid in Linux kernel development
- ❑ Git isn't the only version control system

KEY CONCEPTS: SNAPSHOTS

- ❑ The way git keeps track of your code history
- ❑ Essentially records what all your files look like at a given point in time
- ❑ You decide when to take a snapshot, and of what files
- ❑ Have the ability to go back to visit any snapshot
- ❑ Your snapshots from later on will stay around, too

KEY CONCEPTS: COMMIT

- ❑ The act of creating a snapshot
- ❑ Can be a noun or verb
 - ❑ “I committed code”
 - ❑ “I just made a new commit”
- ❑ Essentially, a project is made up of a bunch of commits

KEY CONCEPTS: COMMIT

Commits contain three pieces of information:

- ❑ 1. Information about how the files changed from previously
- ❑ 2. A reference to the commit that came before it
 - ❑ Called the “parent commit”
- ❑ 3. A hash code name
 - ❑ Will look something like:
f2d2ec5069fc6776c80b3ad6b7cbde3cade4e

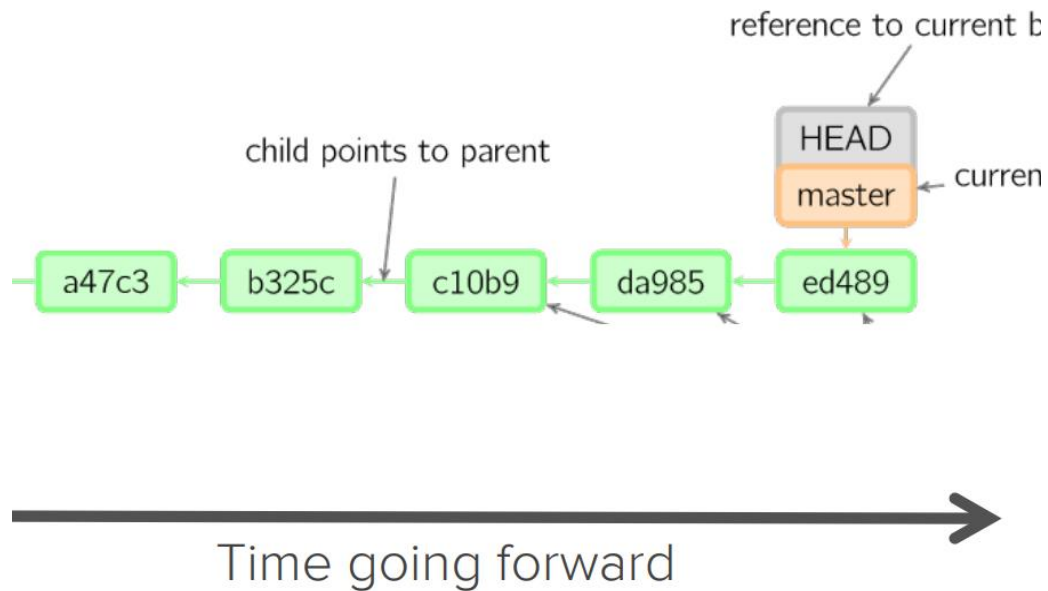
KEY CONCEPTS: REPOSITORIES

- ❑ Often shortened to 'repo'
- ❑ A collection of all the files and the history of those files
- ❑ Consists of all your commits
- ❑ Place where all your hard work is stored

KEY CONCEPTS: REPOSITORIES

- ❑ Can live on a local machine or on a remote server (GitHub!)
- ❑ The act of copying a repository from a remote server is called cloning
- ❑ Cloning from a remote server allows teams to work together
- ❑ The process of downloading commits that don't exist on your machine from a remote repository is called pulling changes
- ❑ The process of adding your local changes to the remote repository is called pushing changes

KEY CONCEPTS: BRANCHES

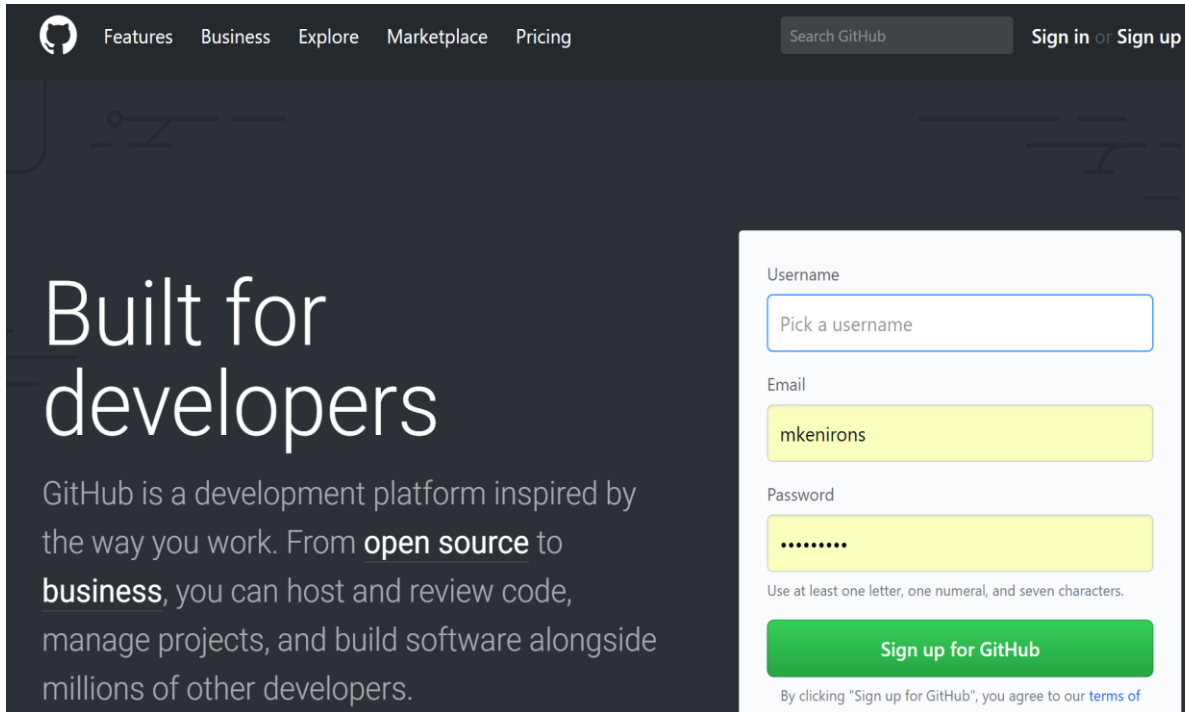


- ❑ All commits in git live on some branch
- ❑ But there can be many, many branches
- ❑ The main branch in a project is called the master branch

WHAT IS GITHUB?

- ❑ Largest web-based git repository hosting service
 - ❑ Aka, hosts 'remote repositories'
- ❑ Allows for code collaboration with anyone online
- ❑ Adds extra functionality on top of git
 - ❑ UI, documentation, bug tracking, feature requests, pull requests, and more!

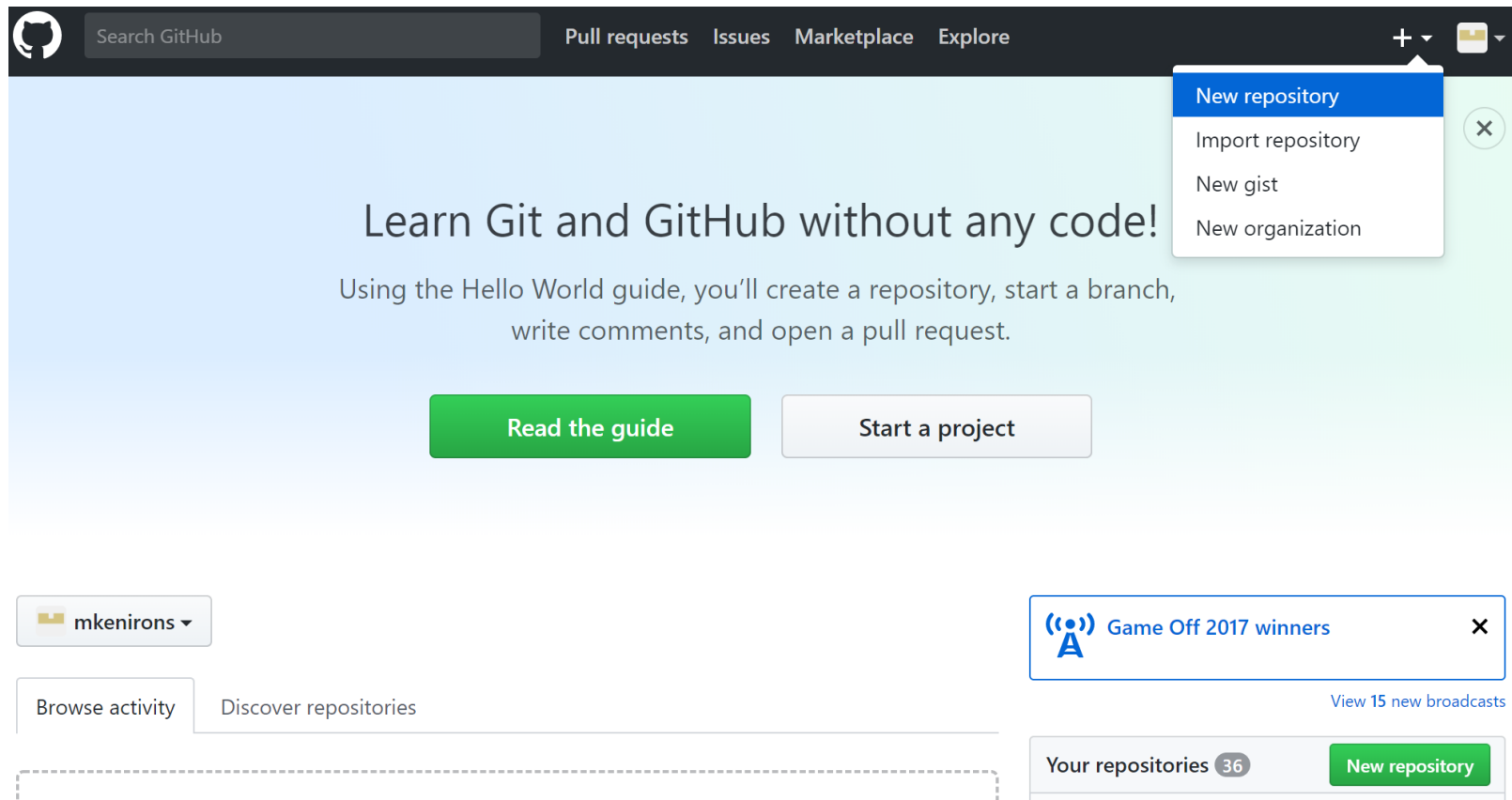
USING GITHUB IN TODAYS LAB



- Create a GitHub Account
 - www.github.com
 - Free Public Repositories

REMOTE REPOSITORY: GITHUB

STEP ONE: CREATE A REPOSITORY ON GITHUB



The screenshot shows the GitHub homepage. At the top, there's a dark navigation bar with the GitHub logo, a search bar labeled 'Search GitHub', and links for 'Pull requests', 'Issues', 'Marketplace', and 'Explore'. On the right of the navigation bar, there's a '+' icon and a folder icon. The '+' icon's dropdown menu is open, showing options: 'New repository' (highlighted in blue), 'Import repository', 'New gist', and 'New organization'. The main content area has a light blue background with the text 'Learn Git and GitHub without any code!' and a subtext 'Using the Hello World guide, you'll create a repository, start a branch, write comments, and open a pull request.' Below this are two buttons: 'Read the guide' (green) and 'Start a project' (light gray). At the bottom, there's a user profile section for 'mkenirons' with a dropdown arrow. To the right, there's a notification for 'Game Off 2017 winners' with a close button. Below the notification, it says 'View 15 new broadcasts'. At the bottom right, there's a section for 'Your repositories' showing '36' repositories and a 'New repository' button.

Search GitHub Pull requests Issues Marketplace Explore

Learn Git and GitHub without any code!

Using the Hello World guide, you'll create a repository, start a branch, write comments, and open a pull request.

Read the guide Start a project

mkenirons

Browse activity Discover repositories

Game Off 2017 winners

View 15 new broadcasts

Your repositories 36 New repository

STEP ONE: CREATE A REPOSITORY ON GITHUB

 Search GitHub

[Pull requests](#) [Issues](#) [Marketplace](#) [Explore](#)

Create a new repository

A repository contains all the files for your project, including the revision history.

Owner

Repository name

 mkenirons

 /

TypeScript 

Great repository names are short and memorable. Need inspiration? How about [supreme-succotash](#).

Description (optional)

☒  **Public**

Anyone can see this repository. You choose who can commit.

☐  **Private**

You choose who can see and commit to this repository.

☐ **Initialize this repository with a README**

This will let you immediately clone the repository to your computer. Skip this step if you're importing an existing repository.

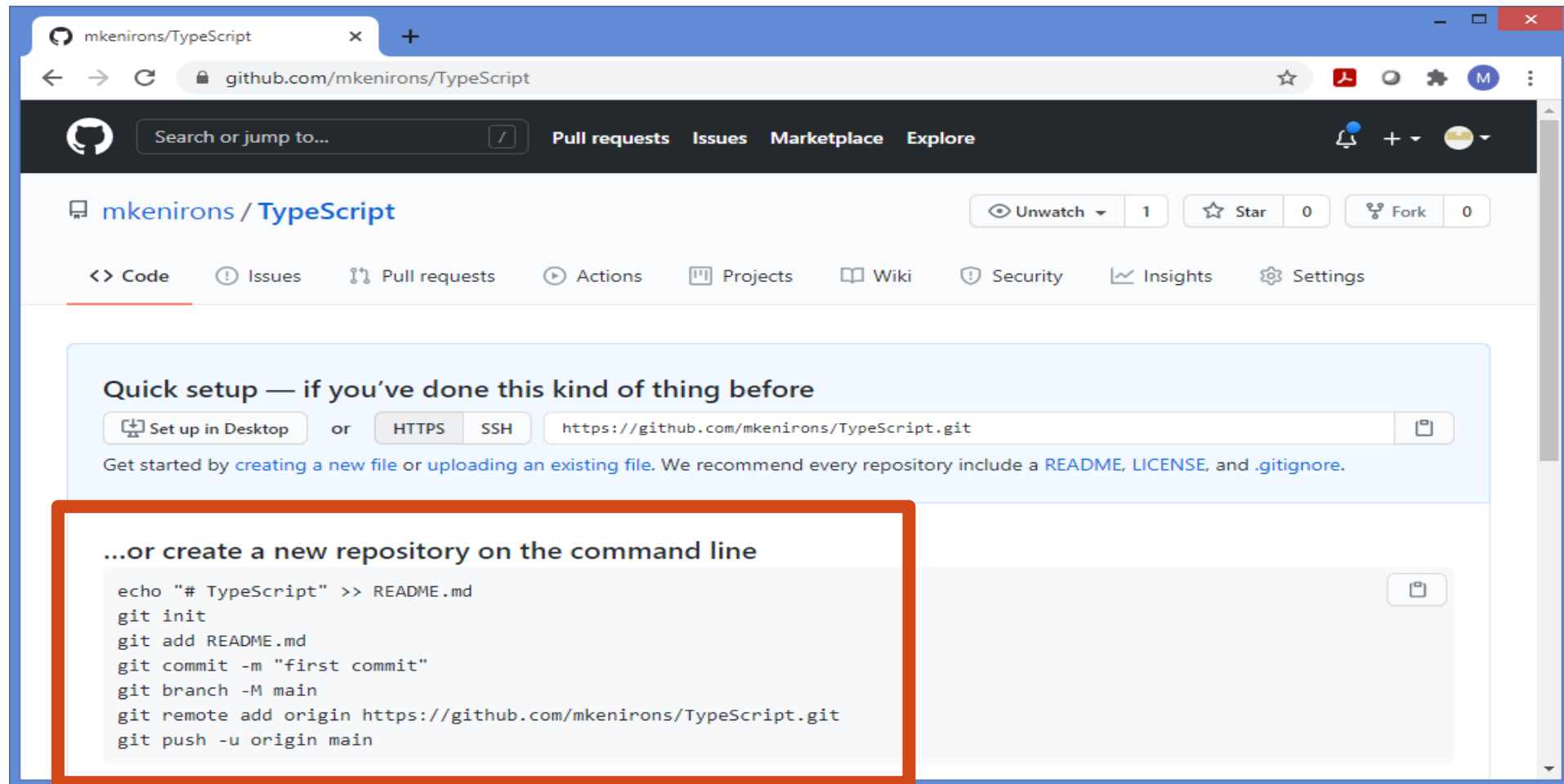
Add .gitignore: None

 |

Add a license: None 

Create repository

STEP TWO: SAVE GIT COMMANDS FROM BLANK REPOSITORY



The screenshot shows the GitHub repository page for `mkenirons/TypeScript`. The page includes a search bar, navigation links for Pull requests, Issues, Marketplace, and Explore, and repository statistics (1 star, 0 forks). The 'Code' tab is selected, displaying the 'Quick setup' section. This section offers options to 'Set up in Desktop', 'HTTPS', or 'SSH', along with the repository URL: `https://github.com/mkenirons/TypeScript.git`. Below this, a section titled '...or create a new repository on the command line' is highlighted with a red box. It contains a list of Git commands to initialize a new repository, create a README, and push to GitHub.

Quick setup — if you've done this kind of thing before

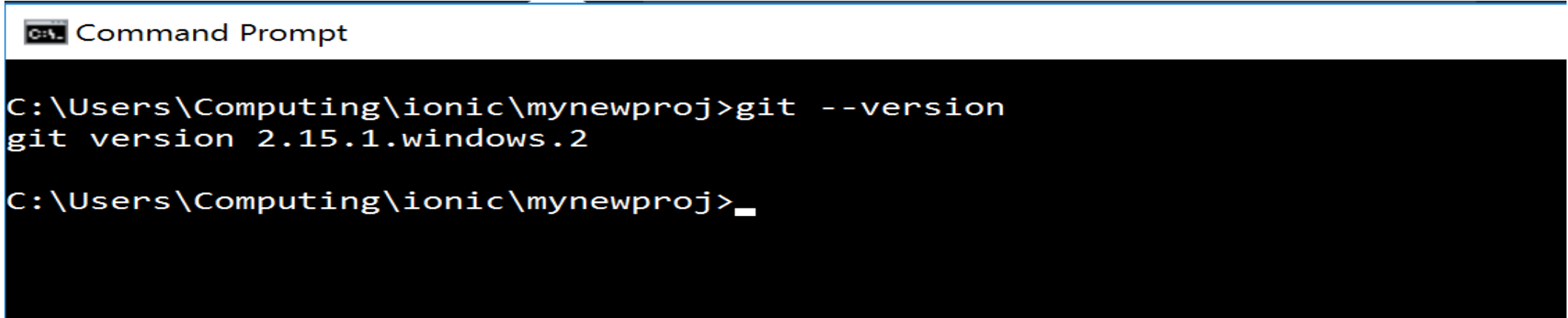
`Set up in Desktop` or `HTTPS` `SSH` `https://github.com/mkenirons/TypeScript.git`

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# TypeScript" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/mkenirons/TypeScript.git
git push -u origin main
```

STEP THREE: INSTALL GIT ON YOUR MACHINE FROM [HTTP://GITFORWINDOWS.ORG/](http://gitforwindows.org/)



```
C:\ Command Prompt

C:\Users\Computing\ionic\mynewproj>git --version
git version 2.15.1.windows.2

C:\Users\Computing\ionic\mynewproj>_
```

STEP FOUR: ADD GIT TO YOUR PATH (ENVIRONMENT VARIABLE)

STEP FIVE: OPEN A WINDOWS COMMAND PROMPT: RUN THE ABOVE COMMAND

STEP SIX: CREATE A FOLDER ON YOUR COMPUTER TO COMPLETE YOUR LAB

STEP SEVEN: CREATE README.MD

STEP EIGHT: ADD THE GIT COOMANDS TO THE README.MD FILE

SREP NINE: CREATE A LOCAL REPOSITORY ON YOUR MACHINE

```
Command Prompt
C:\Users\Computing>mkdir typescript_lab_one
C:\Users\Computing>cd typescript_lab_one
C:\Users\Computing\typescript_lab_one>echo "# TypeScript" >> README.md
C:\Users\Computing\typescript_lab_one>dir
Volume in drive C is Local Disk
Volume Serial Number is 0671-FAEA

Directory of C:\Users\Computing\typescript_lab_one

21/01/2018  17:43    <DIR>          .
21/01/2018  17:43    <DIR>          ..
21/01/2018  17:43                17 README.md
               1 File(s)                17 bytes
               2 Dir(s)  177,523,277,824 bytes free

C:\Users\Computing\typescript_lab_one>git init
Initialized empty Git repository in C:/Users/Computing/typescript_lab_one/.git/
C:\Users\Computing\typescript_lab_one>
```

```
README.MD X
C: > Users > Martin > README.MD
1  echo "# TypeScript" >> README.md
2  git init
3  git add README.md
4  git commit -m "first commit"
5  git branch -M main
6  git remote add origin https://github.com/mkenirons/TypeScript.git
7  git push -u origin main
```

STEP 10: ADD ALL FILES FROM OUR FOLDER FOR YOUR LOCAL REPOSITORY

STEP 11: COMMIT THESE FILE TO THE LOCAL REPOSITORY

STEP 12: MAKE A NEW BRANCH CALLED MAIN

STEP 13: LINK THE LOCAL REPOSITORY TO THE REMOTE REPOSITORY

STEP 14: PUSH FILES FROM LOCAL TO REMOTE REPOSITORY

```
C:\Users\Computing\typescript_lab_one>git init
Initialized empty Git repository in C:/Users/Computing/typescript_lab_one/.git/

C:\Users\Computing\typescript_lab_one>git add .      10

C:\Users\Computing\typescript_lab_one>git commit -m "Some Comment about our first commit"
[master (root-commit) 9f29912] Some Comment about our first commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md      11

C:\Users\Computing\typescript_lab_one>git branch -M main      12

C:\Users\Computing\typescript_lab_one>git remote add origin https://github.com/mkenirons/TypeScript.git
      13

C:\Users\Computing\typescript_lab_one>git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 243 bytes | 243.00 KiB/s, done.      14
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/mkenirons/TypeScript.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.

C:\Users\Computing\typescript_lab_one>
```

STEP 15: AFTER CREATING MORE FILES — AGAIN ADD ALL FILES TO LOCAL REPO

STEP 16: COMMIT THESE FILE TO THE LOCAL REPOSITORY

STEP 17: PUSH FILES FROM LOCAL TO REMOTE REPOSITORY

```
G:\Users\Computing\typescript_lab_one>git add . 15
G:\Users\Computing\typescript_lab_one>git commit -m "Some Comment about our second commit"
[main e29ce61] Some Comment about our second commit 16
 1 file changed, 1 insertion(+)
 create mode 100644 types.ts
G:\Users\Computing\typescript_lab_one>git push -u origin main 17
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 300 bytes | 300.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/mkenirons/TypeScript.git
 9f29912..e29ce61  main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
G:\Users\Computing\typescript_lab_one>
```

The screenshot shows the GitHub interface for the repository 'mkenirons / TypeScript'. The repository has 0 stars, 0 forks, and 0 issues. The commit history shows two commits: 'some comment about our first commit' (12 minutes ago) and 'some comment about our second commit' (3 minutes ago). The file list includes 'README.md' and 'types.ts'. The 'types.ts' file is highlighted, showing its content: 'some comment about our second commit'.

This repository Search Pull requests Issues Marketplace Explore

mkenirons / TypeScript Watch 0 Star 0 Fork 0

Code Issues 0 Pull requests 0 Projects 0 Wiki Insights Settings

description, website, or topics provided. Edit

2 commits 1 branch 0 releases 1 contributor

h: master New pull request Create new file Upload files Find file Clone or download

mkenirons some comment about our second commit Latest commit 09aa391 3 minutes ago

README.md	some comment about our first commit	12 minutes ago
types.ts	some comment about our second commit	3 minutes ago

README.md

LAST SLIDE